

Relational Database Architecture & ERD Specification

Production-Ready Data Architecture, PostGIS Geospatial Engine & Anti-Scam Governance

Jira Reference: SCRUM-53

Database Engine: PostgreSQL 15+ (Supabase)

Status: **APPROVED / DONE**

Lead / Scrum Master: Hermar Centillas

Extensions: postgis, uuid-oss, moddatetime

Version: 1.0.0 (Production)

GitHub Repository Source: github.com/hermarDev/Abang_Cebu_AI_Group2/blob/main/docs/database-erd.md

1. Executive Summary & Design Pillars

The AbangCebuAI database is architected to power an intelligent, map-centric rental marketplace tailored specifically for Metro Cebu (Cebu City, Mandaue, Lapu-Lapu, Talisay, and surrounding towns). The database satisfies four foundational criteria:

- Third Normal Form (3NF) & Integrity:** Complete normalization across 11 core domain tables, avoiding data duplication while establishing strict foreign key referential integrity with automated delete cascades.
- PostGIS Geospatial Engine:** WGS84 coordinates stored as native geometry(Point, 4326) accompanied by GIST spatial indexing to enable sub-50ms MapLibre bounding box queries and proximity radius searches.
- Real Estate Unit Hierarchy:** Decouples physical parent buildings (properties) from leasable components (rental_units), cleanly supporting condominiums, apartments, private rooms, and bedspace boarding houses.
- Anti-Scam Trust Layer:** Mandatory landlord KYC verification workflow (kyc_verifications) requiring government ID validation and property title/bill proofs before listings can be approved for public viewing.

Traceability & Governance: This document and diagram serve as the authoritative schema standard for Sprint 1. It directly unblocks SCRUM-54 (Users Schema), SCRUM-55 (RLS Policies), and MapLibre search query implementations.

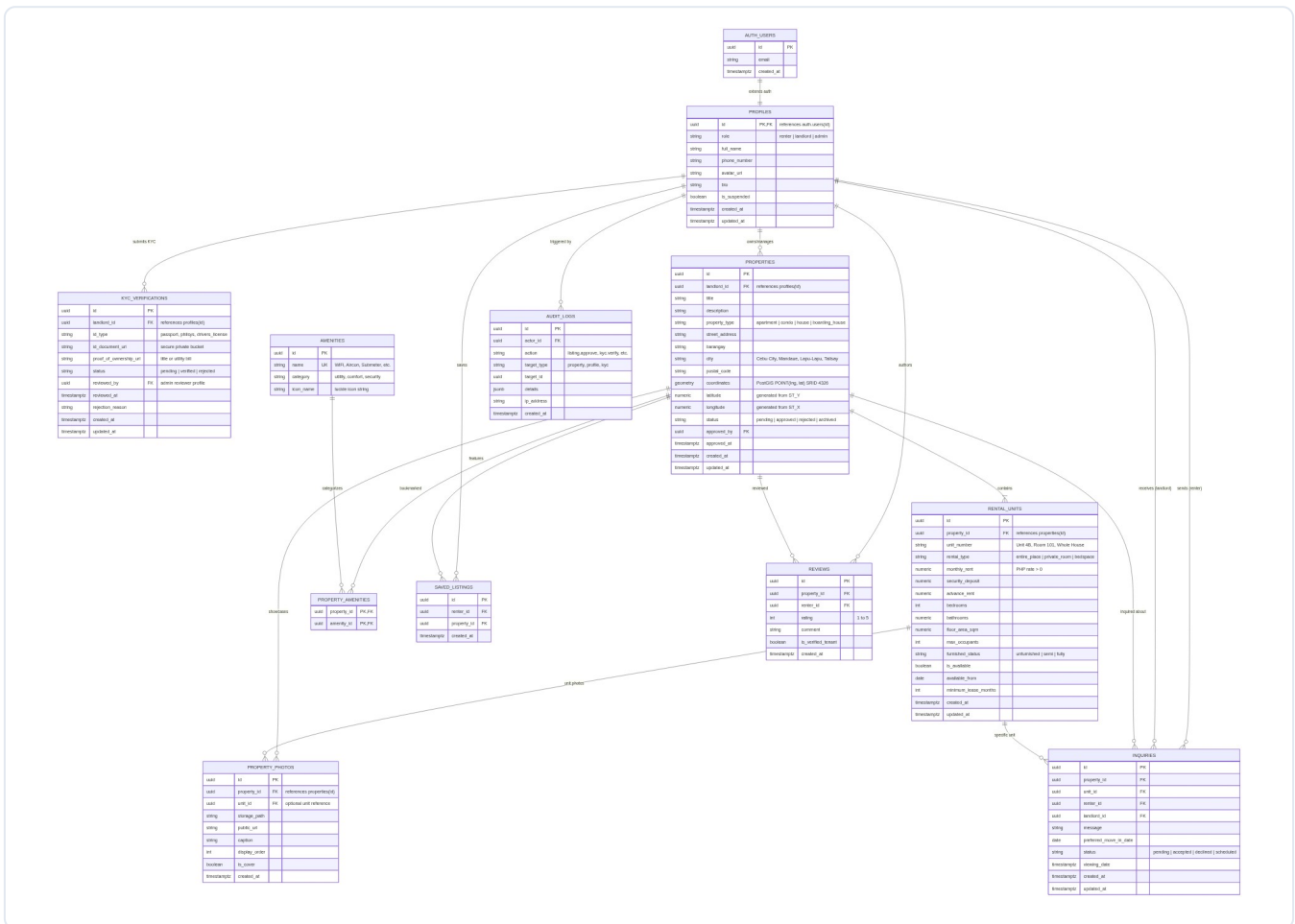
2. Core Entities Overview

Entity	Domain	Description & Role in System
profiles	Identity / Auth	Extends auth.users with user role (renter, landlord, admin), display details, and suspension flags.
kyc_verifications	Anti-Scam Trust	Manages government ID uploads, proof of ownership documents, and admin verification status.
properties	Real Estate / GIS	Physical property buildings with PostGIS WGS84 coordinates, barangay, and city location metadata.
rental_units	Pricing & Inventory	Leasable units, monthly rent (PHP), deposits, bedrooms, bathrooms, and availability dates.
property_photos	Media Attachments	Cloud storage paths, public CDN URLs, display order, and thumbnail cover flags.
amenities / property_amenities	Features & Utilities	Normalized many-to-many catalog of inclusions: WiFi, Aircon, Sub-meters, Security, Parking, etc.
inquiries	Communication	Lead messages, preferred move-in dates, and scheduled property viewing appointments.
saved_listings	User Engagement	

Entity	Domain	Description & Role in System
		Renter saved bookmarks and favorite rental listings for quick dashboard retrieval.
reviews	Reputation	Verified tenant ratings (1-5 stars) and reviews evaluating landlord service quality.
audit_logs	Administrative Audit	Immutable system audit trail recording listing approvals, rejections, and user role modifications.

3. Visual Entity-Relationship Diagram (ERD)

The following high-resolution diagram illustrates all 11 entities, relationships, cardinality, and key constraints across the AbangCebuAI database:



Vector SVG version available in repository: docs / assets / abangcebu_database_erd.svg

4. Data Dictionary — Core Tables

4.1 (User Identity & Role Governance)

Column	Type	Nullable	Default	Description & Constraints
id PK FK	uuid	NO	-	Foreign Key referencing auth.users(id) ON DELETE CASCADE.
role	text	NO	'renter'	User role enum: 'renter', 'landlord', 'admin'.
full_name	text	NO	-	Full display name of user.
phone_number	text	YES	NULL	Contact number (Philippine format: +639...). Protected by RBAC.
avatar_url	text	YES	NULL	Storage URL for profile picture.
bio	text	YES	NULL	Biography or landlord profile introduction.
is_suspended	boolean	NO	false	Account suspension flag. If true, user actions are rejected with 403.
created_at / updated_at	timestampz	NO	now()	Audit timestamps with automated moddatetime trigger.

4.2 (Anti-Scam Landlord Identity Verification)

Column	Type	Nullable	Default	Description & Constraints
id PK	uuid	NO	uuid_generate_v4()	Primary Key.
landlord_id FK	uuid	NO	-	Foreign Key referencing profiles(id) ON DELETE CASCADE.
id_type	text	NO	-	Valid ID type: passport, philsys_id, drivers_license, umid.
id_document_url	text	NO	-	Encrypted private storage path to government ID document scan.
proof_of_ownership_url	text	YES	NULL	Private storage path to land title, tax declaration, or utility bill.
status	text	NO	'pending'	Verification status: 'pending', 'verified', 'rejected'.
reviewed_by FK	uuid	YES	NULL	Foreign Key referencing profiles(id) of reviewing Admin.
reviewed_at / rejection_reason	mixed	YES	NULL	Review audit timestamp and feedback note if rejected.

4.3 (Real Estate Buildings & Locations)

Column	Type	Nullable	Default	Description & Constraints
id PK	uuid	NO	uuid_generate_v4()	Primary Key.
landlord_id FK	uuid	NO	-	Foreign Key referencing profiles(id) ON DELETE CASCADE.

Column	Type	Nullable	Default	Description & Constraints
title / description	text	NO	-	Property title and detailed description.
property_type	text	NO	-	apartment, condominium, house, townhouse, commercial, boarding_house.
barangay / city	text	NO	'Cebu City'	Metro Cebu administrative divisions (e.g. Apas, Lahug, Mandaue City).
coordinates PostGIS	geometry(Point, 4326)	NO	-	WGS84 2D Point. Indexed with GIST for spatial search.
latitude / longitude	numeric(10, 7)	NO	Generated	Stored generated columns via ST_Y(coordinates) and ST_X(coordinates).
status	text	NO	'pending'	Lifecycle: 'pending', 'approved', 'rejected', 'archived'.

4.4 (Granular Inventory & Pricing)

Column	Type	Nullable	Default	Description & Constraints
id PK	uuid	NO	uuid_generate_v4()	Primary Key.
property_id FK	uuid	NO	-	Foreign Key referencing properties(id) ON DELETE CASCADE.
rental_type	text	NO	-	entire_place, private_room, shared_room, bedspace.
monthly_rent	numeric(10, 2)	NO	-	Monthly rent rate in Philippine Pesos (CHECK (monthly_rent > 0)).
security_deposit / advance_rent	numeric(10, 2)	NO	0.00	Required upfront deposit terms.
bedrooms / bathrooms	int / numeric(3,1)	NO	1 / 1.0	Room counts (bathrooms supports .5 for powder rooms).
furnished_status	text	NO	'unfurnished'	unfurnished, semi_furnished, fully_furnished.
is_available / available_from	bool / date	NO	true / CURRENT_DATE	Current lease availability flag and move-in date.
minimum_lease_months	int	NO	6	Minimum contract duration in months.

5. PostGIS Geospatial Engine & Query Optimization

All property coordinates are stored using WGS 84 (SRID 4326). A Generalized Search Tree (GIST) index accelerates spatial indexing:

```
CREATE INDEX idx_properties_coordinates ON properties USING GIST (coordinates);
```

Query Pattern A: MapLibre Viewport Bounding Box (Map Panning)

```
SELECT p.id, p.title, p.latitude, p.longitude, MIN(u.monthly_rent) as starting_price
FROM properties p
JOIN rental_units u ON u.property_id = p.id
WHERE p.status = 'approved'
  AND p.coordinates && extensions.ST_MakeEnvelope(
    123.8800, 10.3100, -- minLng, minLat (South-West)
    123.9200, 10.3500, -- maxLng, maxLat (North-East)
    4326
  )
GROUP BY p.id;
```

Query Pattern B: Landmark Radius Search (e.g. 2km from Cebu IT Park)

```
SELECT p.id, p.title, p.city, p.barangay,
  extensions.ST_Distance(
    p.coordinates::extensions.geography,
    extensions.ST_MakePoint(123.9065, 10.3298)::extensions.geography
  ) as distance_meters
FROM properties p
WHERE p.status = 'approved'
  AND extensions.ST_DWithin(
    p.coordinates::extensions.geography,
    extensions.ST_MakePoint(123.9065, 10.3298)::extensions.geography,
    2000 -- 2 kilometers radius
```

```
)  
ORDER BY distance_meters ASC;
```

6. Architectural Sign-Off & Handover

The schema and ERD specification satisfy 100% of Sprint 1 foundational criteria:

- ☒ All 11 domain entities fully specified in 3NF normalization.
- ☒ Zero premature application/UI code committed (pure architectural foundation).
- ☒ Full alignment with RBAC Matrix (SCRUM-63) and Landlord Persona (SCRUM-62).
- ☒ Official documentation archived in repository: docs/database-erd.md.